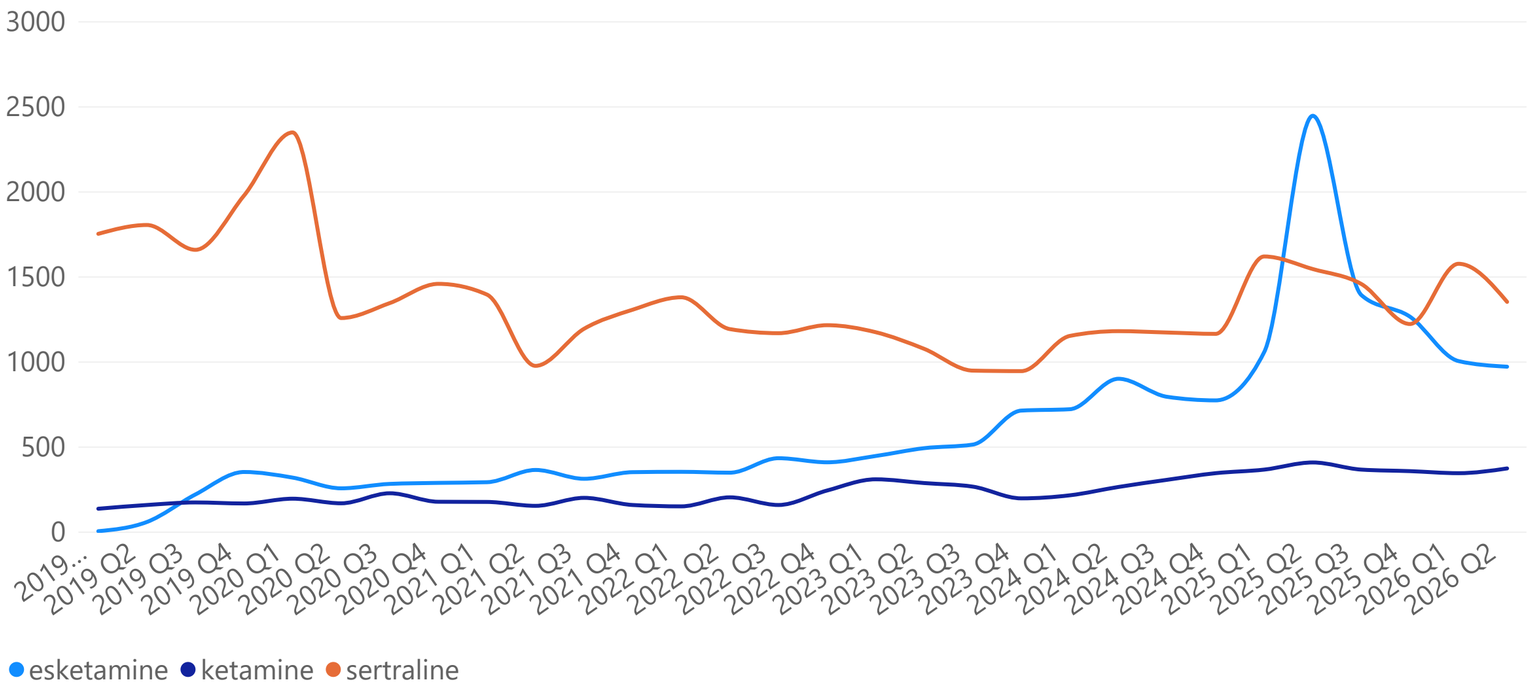
Esketamine and ketamine adverse event reports, FAERS 2019 Q1 to 2026 Q2

These are spontaneous reports. A signal means a reaction is reported more often with a drug than with other drugs; it does not show how often the reaction happens or that the drug caused it.

FAERS cases	Esketamine reports	Ketamine reports	Positive controls passed
10,825,833	18,068	7186	8 of 8

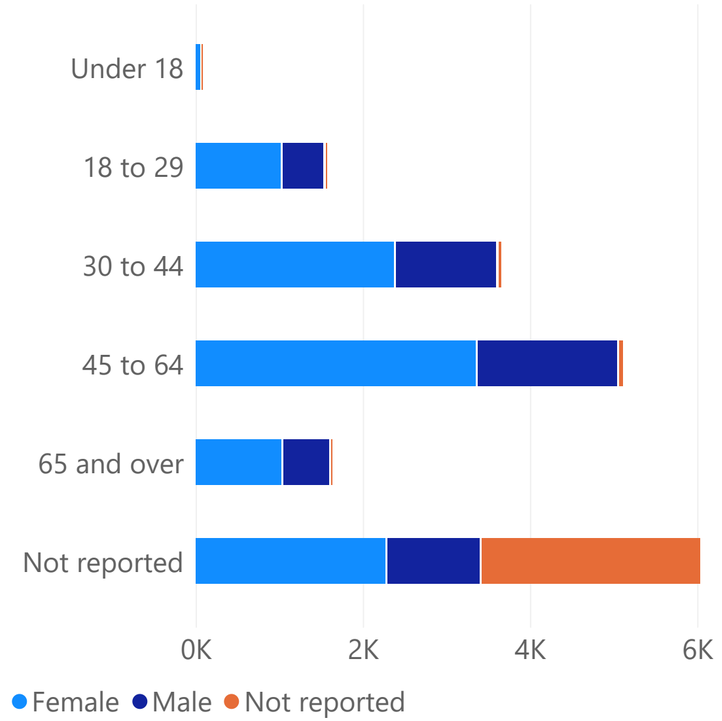
FAERS reports per quarter: esketamine, ketamine and sertraline for reference

Esketamine reports reached 2,444 in 2025 Q2, about twice the usual quarter. A burst of reports is a change in reporting, not evidence of a change in risk.



Esketamine reports by age band and sex

Reports by Age band and Sex



Signal finder: reactions reported more often than expected for one drug

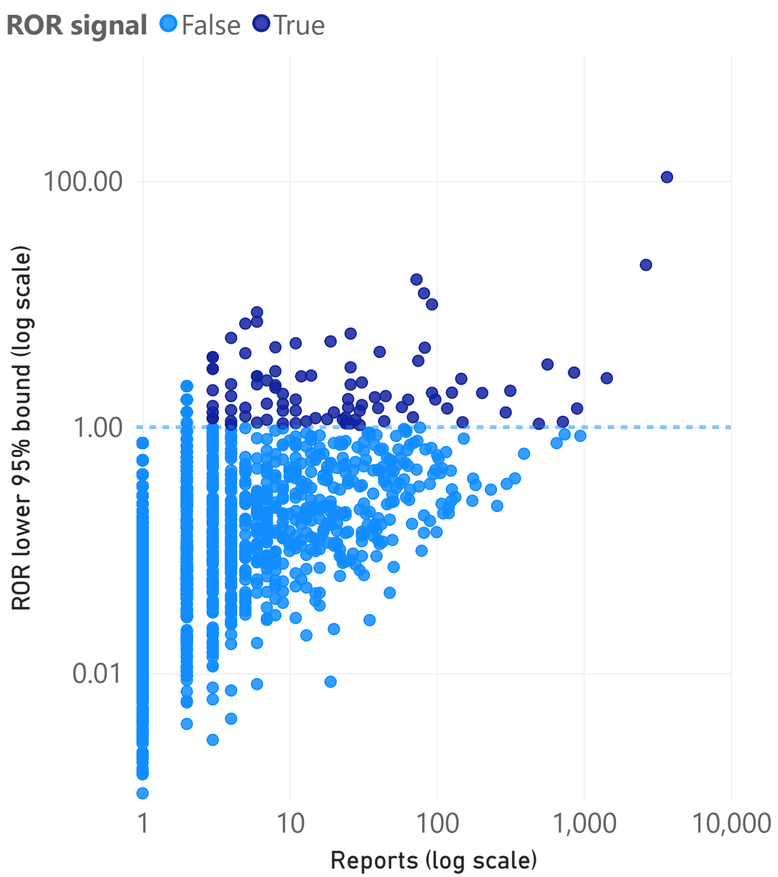
Drug: esketamine

Compared with: Other antidepressants

Term type: Clinical

Reaction terms for the selected drug: reports and lower bound of the ROR

Each point is one term. Points above the line at 1 have a lower 95% bound above 1. Both axes use a log scale.



Signals for the selected drug, largest lower bound first

A signal has at least 3 reports and a lower 95% bound above 1. Only esketamine and ketamine have the Other antidepressants comparison; for any other drug, set Compared with to All reports.

Reaction	Organ class	Reports	ROR	ROR lower 95%	ROR upper 95%	PRR	Evans signal
Dissociation	Psychiatric disorders	3,691	118.70	107.61	130.94	94.66	True
Sedation	Nervous system disorders	2,654	22.17	20.81	23.63	19.06	True
Nasal discomfort	Respiratory, thoracic and mediastinal disorders	73	23.52	15.83	34.94	23.43	True
Oxygen saturation abnormal	Investigations	82	17.15	12.23	24.06	17.08	True
Dissociative disorder	Psychiatric disorders	93	13.37	9.94	17.98	13.30	True
Pregnancy of partner	Social circumstances	6	71.27	8.58	592.01	71.24	True
Living in residential institution	Social circumstances	6	35.63	7.19	176.56	35.62	True
Nasal septum perforation	Respiratory, thoracic and mediastinal disorders	5	59.39	6.94	508.35	59.37	True
Psychogenic seizure	Nervous system disorders	26	9.66	5.76	16.21	9.65	True
Vagal nerve stimulator implantation	Surgical and medical procedures	4	47.51	5.31	425.07	47.50	True
Flashback	Psychiatric disorders	19	9.03	4.97	16.40	9.02	True

Drug comparison: ROR signals by organ class

Right-click any cell and choose Drill through, then Terms behind a cell, to list the reaction terms behind it. Each cell counts clinical terms flagged against all other FAERS reports.

Esketamine, ketamine and SSRIs: number of ROR signals by organ class

Clinical terms, compared with all other FAERS reports.

Organ class	citalopram	escitalopram	esketamine	fluoxetine	ketamine	paroxetine	sertraline
Blood and lymphatic system disorders	14	25		17	7	20	19
Cardiac disorders	60	61	1	55	36	45	69
Congenital, familial and genetic disorders	46	45		43	7	56	71
Ear and labyrinth disorders	12	7	3	6	1	4	16
Endocrine disorders	9	6		10	2	4	15
Eye disorders	18	47	3	29	10	31	53
Gastrointestinal disorders	71	73	3	49	26	49	102
General disorders and administration site conditions	57	69	8	55	31	48	64
Hepatobiliary disorders	11	11		13	29	22	21
Immune system disorders	7	13	1	8	5	9	12
Infections and infestations	35	22	2	29	19	27	27
Injury, poisoning and procedural complications	51	44	10	38	28	46	51
Investigations	101	118	6	114	35	76	116
Metabolism and nutrition disorders	34	47		36	20	34	43
Musculoskeletal and connective tissue disorders	38	34		33	6	37	41
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	7	14		9		7	11
Nervous system disorders	151	185	32	148	71	151	207
Not mapped	63	61	2	68	20	50	130
Pregnancy, puerperium and perinatal conditions	15	24		30	3	21	34
Psychiatric disorders	170	213	95	200	68	168	239
Renal and urinary disorders	20	31	8	18	23	25	32
Reproductive system and breast disorders	34	49		39		22	61
Respiratory, thoracic and mediastinal disorders	70	71	11	55	37	46	78
Skin and subcutaneous tissue disorders	27	30	1	23	20	36	35
Social circumstances	16	29	5	18	17	25	26
Surgical and medical procedures	12	11	5	6	7	4	7
Vascular disorders	22	27	5	22	19	26	25
Total	**1171**	**1367**	**201**	**1171**	**547**	**1089**	**1605**

SNRIs and other antidepressants: number of ROR signals by organ class

Clinical terms, compared with all other FAERS reports.

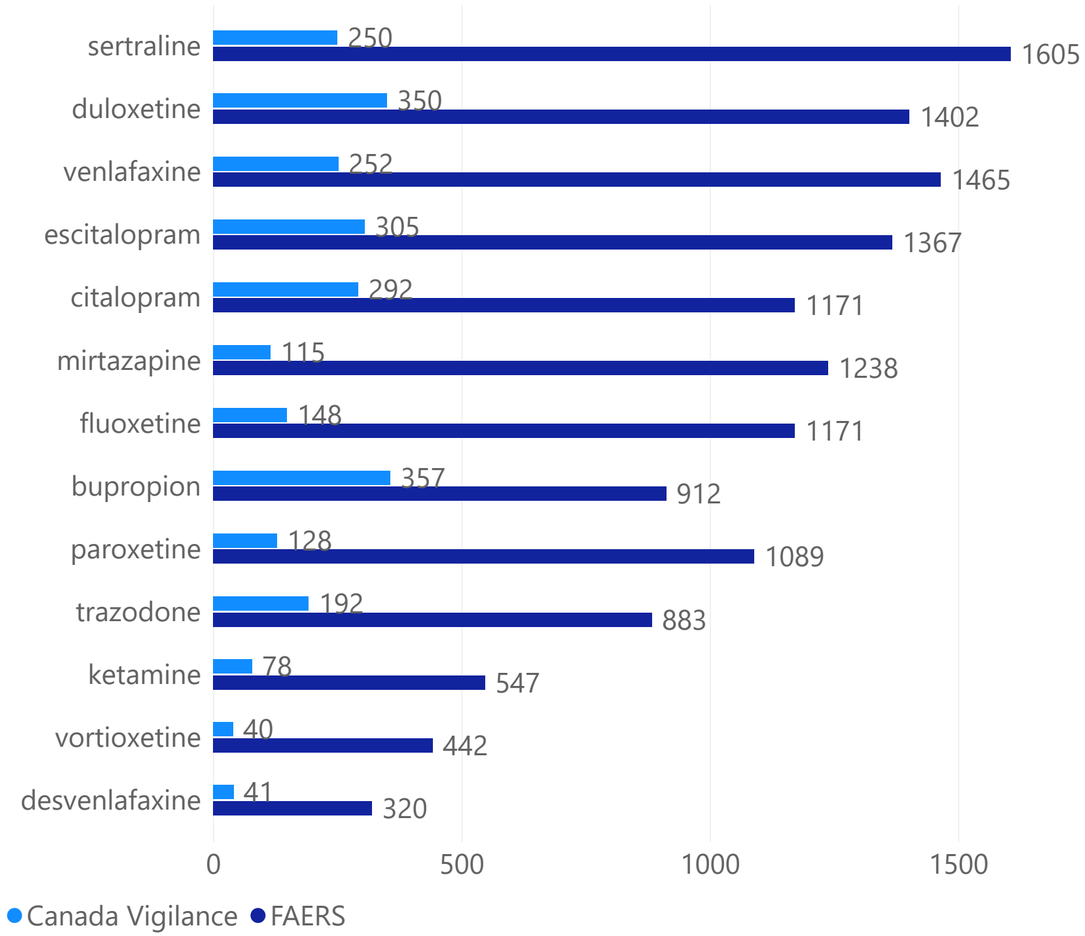
Organ class	bupropion	desvenlafaxine	duloxetine	mirtazapine	trazodone	venlafaxine	vortioxetine
Blood and lymphatic system disorders	5		16	17	7	20	4
Cardiac disorders	45	6	50	50	48	82	12
Congenital, familial and genetic disorders	20	2	37	37	8	46	2
Ear and labyrinth disorders	5	8	10	10	3	13	4
Endocrine disorders	8	2	12	9	6	6	4
Eye disorders	14	10	37	29	15	44	19
Gastrointestinal disorders	53	14	82	62	46	70	28
General disorders and administration site conditions	48	24	79	58	58	65	24
Hepatobiliary disorders	6	2	24	19	15	27	4
Immune system disorders	15	5	12	4	7	11	1
Infections and infestations	16	2	36	29	30	26	6
Injury, poisoning and procedural complications	35	16	68	49	42	51	18
Investigations	78	19	129	122	96	139	25
Metabolism and nutrition disorders	26	7	40	43	32	45	14
Musculoskeletal and connective tissue disorders	23	6	93	50	41	42	8
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	5	1	9	11	10	18	
Nervous system disorders	130	55	197	175	114	191	73
Not mapped	30	9	45	52	20	91	11
Pregnancy, puerperium and perinatal conditions	14	2	20	20	10	26	3
Psychiatric disorders	185	100	192	201	137	215	128
Renal and urinary disorders	14	2	32	31	8	31	8
Reproductive system and breast disorders	23	4	30	38	13	33	17
Respiratory, thoracic and mediastinal disorders	41	8	73	56	49	77	7
Skin and subcutaneous tissue disorders	35	6	27	15	25	29	6
Social circumstances	13	7	18	20	21	32	5
Surgical and medical procedures	7		6	4	6	10	3
Vascular disorders	18	3	28	27	16	25	8
Total	**912**	**320**	**1402**	**1238**	**883**	**1465**	**442**

Canada Vigilance compared with FAERS

Canada Vigilance holds 19 esketamine reports received from 2019 to June 2026, against 18,068 in FAERS, too few for disproportionality analysis, so this page compares ketamine and the 12 other antidepressants.

Clinical ROR signals per drug, FAERS and Canada Vigilance

Canada Vigilance has about 5% as many reports as FAERS, so its confidence intervals are wider and it flags fewer signals.



Drug (table only)

sertraline

Top 15 reactions by FAERS report count for the selected drug

A blank Canada cell means the reaction was never reported for that drug in Canada.

Reaction	FAERS reports	FAERS ROR lower 95%	Canada reports	Canada ROR lower 95%
Intentional overdose	2237	17.25	11	9.16
Nausea	2138	1.49	94	1.48
Toxicity to various agents	2094	6.22	46	7.28
Anxiety	1921	3.87	77	4.10
Dizziness	1806	2.08	62	1.51
Headache	1759	1.54	55	0.93
Drug interaction	1672	6.08	123	34.49
Fatigue	1596	1.00	49	0.53
Diarrhoea	1560	1.16	85	1.48
Suicide attempt	1494	15.31	18	8.78
Vomiting	1478	1.73	44	1.11
Suicidal ideation	1457	10.52	47	9.24
Somnolence	1421	3.96	23	1.70
Insomnia	1414	3.26	37	1.78
Completed suicide	1363	11.68	10	11.66

Methods, validation and data

Data

FAERS quarterly files from 2019 Q1 to 2026 Q2: 12,902,877 report versions, reduced to 10,825,833 cases by keeping the latest version of each case and removing cases the FDA deleted. Canada Vigilance extract dated 30 June 2026: 567,733 reports received from 2019 onward after removing 26,988 linked duplicates. Only drugs listed as suspect count.

Method

Each drug and reaction gets a 2 x 2 table of report counts. a = reports with the drug and the reaction; b = reports with the drug and any other reaction; c = other reports with the reaction; d = other reports without it. ROR = (a/b) / (c/d). PRR = (a/(a+b)) / (c/(c+d)).

The comparison group is every other report in the database. Esketamine and ketamine are also compared with reports for 12 other antidepressants. A signal is a reaction with at least 3 reports and a lower 95% confidence bound for the reporting odds ratio above 1. The proportional reporting ratio with the Evans criteria is shown for comparison. Reaction terms that describe products, devices, procedures or lack of effect are left out of the clinical results; the Term type slicer on the Signal finder page shows them.

Limitations

Spontaneous reports can show that a reaction is reported more often with a drug. They cannot show how often it happens or that the drug caused it, and there is no count of people exposed to each drug.

People prescribed esketamine have treatment-resistant depression, so psychiatric reactions such as suicidal ideation are expected to be reported more often. Spravato is given in certified clinics with required monitoring, which likely raises its reporting compared with antidepressants taken at home. Esketamine reports rose sharply in 2025 Q2. Most ketamine reports come from anaesthesia and sedation; only 251 FAERS ketamine reports name a psychiatric use. Organ classes come from Canada Vigilance's MedDRA coding and cover 95% of FAERS drug-reaction pairs.

Validation: known reactions (should flag) and unrelated reactions (should not)

Drug	Reaction	Expected	Flagged	Reports	ROR	ROR lower 95%	Source
esketamine	Dissociation	True	True	3691	1819.20	1710.06	Spravato label: boxed warning
esketamine	Sedation / Somnolence	True	True	2888	180.60	172.57	Spravato label: boxed warning
esketamine	Suicidal ideation	True	True	1437	26.02	24.63	Spravato label: boxed warning (also confounded by indication)
esketamine	Dizziness	True	True	655	1.77	1.64	Spravato label: common adverse reaction
esketamine	Blood pressure increased	True	True	568	4.56	4.19	Spravato label: warnings
ketamine	Cystitis / Cystitis interstitial / Cystitis noninfective / Cystitis haemorrhagic / Cystitis ulcerative	True	True	163	753.49	304.03	Ketamine-induced uropathy
esketamine	Vertigo	True	True	103	2.36	1.94	Spravato label: common adverse reaction
esketamine	Feeling drunk	True	True	75	17.46	13.87	Spravato label: common adverse reaction
esketamine	Cataract	False	False	17	0.32	0.20	Negative control
esketamine	Diabetes mellitus	False	False	16	0.31	0.19	Negative control
esketamine	Cholelithiasis	False	False	13	0.63	0.37	Negative control
esketamine	Rotator cuff syndrome	False	False	5	0.56	0.23	Negative control

FAERS files loaded, by quarter

Quarter	Report versions	Lines skipped	Cases kept
2019Q1	413734	0	336,806
2019Q2	441108	0	353,206
2019Q3	452873	0	369,180
2019Q4	419581	0	347,660
2020Q1	460327	0	380,189
2020Q2	429227	0	348,212
2020Q3	431667	0	348,632
2020Q4	436148	0	346,643
2021Q1	463741	0	377,474
2021Q2	479945	0	395,866
2021Q3	504160	0	423,386
2021Q4	412542	0	364,661
2022Q1	461623	0	391,655
2022Q2	435618	0	356,486
2022Q3	446511	0	379,725
2022Q4	483643	0	399,048
2023Q1	432144	0	354,257
2023Q2	418592	0	344,068
2023Q3	407522	0	347,693
2023Q4	415379	0	339,470
2024Q1	406184	0	334,949
2024Q2	397119	0	328,498
2024Q3	405513	0	342,974
2024Q4	410849	0	333,635
2025Q1	400514	0	325,021
2025Q2	393130	0	326,905
2025Q3	438512	0	391,621
2025Q4	385288	0	345,568
2026Q1	397224	0	369,887
2026Q2	422459	0	422,458
Total	**12902877**	**0**	**10,825,833**